



Figure 1: The '+' indicates the presence of ACL

included with the mount request, or set in the superblock.

Viewing and interpreting ACL permissions

The `ls-l` command only outputs minimal setting details.

The + at the end of a 10-character permission string indicates that there are ACL settings associated with this file. You can interpret the *user*, *group* and *other* *rw*x flags as follows.

- **User:** Shows the user ACL settings, which are the same as standard user file settings; *rw*x.
- **Group:** Shows the current ACL mask settings, not the group owner settings; *rw*.
- **Other:** Shows the other ACL settings, which are the same as other standard file settings; *no access*.

Note: Changing group permissions on a file with an ACL by using `chmod` does not change the group owner permissions and does not change the ACL mask. Use the `setfacl -m g::perms` file if the intent is to update the file group owner permissions.

Viewing the ACLs

To display ACL settings on a file, use `getfacl file name` (Figure 2).

Each section of the example given in Figure 2 indicates the following.

Opening comment entries: The first three lines are comments that identify the file name, owner (student) and group owner (controller). If there are any additional file flags, for example, `setuid` or `setgid`, then a fourth comment line will appear showing which flags are set.

User entries:

1. File owner permission is *rw*x.
2. The named user is *ram* and *abc* permissions are *rw*x.

Group entries:

1. Group owner permissions are *rw*.
2. Named group permissions. One entry for the group *pqr* shows permission to be *rw*x.

Viewing directory ACLs

To display ACL settings on a directory, you can use `getfacl /`

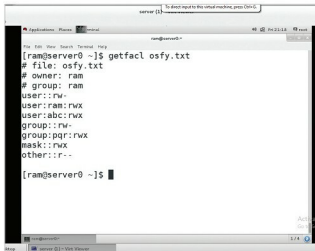


Figure 2: ACL settings on the file *osfy.txt*

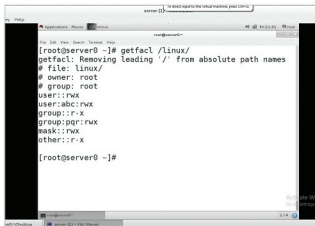


Figure 3: ACL on a directory

directory (Figure 3).

Each section of the example given in Figure 3 indicates the following.

Opening comment entries: The first three lines of comments identify the directory name, the owner (student), and group owner (controller). If there are any additional directory flags (`setuid`, `setgid`, `sticky`), then a fourth comment line will appear showing the set flags—in this case, `setgid`.

Standard ACL entries which are shown below the comment lines are the ACL permissions on this directory. They are the same as the file example mentioned earlier, but apply to the directory. The key difference is the inclusion of the *execute* permission on these entries (when appropriate) to allow directory search permission.

Note: The output from the `getfacl` command can be used as input to `setfacl`. Use `getfacl -R /directory` to generate output for the directory and its contents. This output can be saved and used for recovery by passing the output to `setfacl --set-file=file` to do a mass update.